

## CUSTOMER MILESTONE BRIEF

# MS #50 - Oracle Semantic Coverage Program

Purpose, customer value, delivered capability, evidence, and claim boundaries

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| <b>Status</b>   | Complete                                                                             |
| <b>Release</b>  | 0.50.0, 0.50.1, 0.50.2, 0.50.3, 0.50.4                                               |
| <b>Date</b>     | 2026-09-01                                                                           |
| <b>Audience</b> | Customers, partners, executives, architects, delivery teams, and assurance reviewers |

## Executive summary

MS #50 - Oracle Semantic Coverage Program converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Gives database architects a quantified, evidence-levelled Oracle coverage answer and completes bounded catalog execution across core SQL, datatypes, PL/SQL, transactions, operations, schema/DML, schema objects, and structured data.

## Why this matters to a customer

Gives database architects a quantified, evidence-levelled Oracle coverage answer and completes bounded catalog execution across core SQL, datatypes, PL/SQL, transactions, operations, schema/DML, schema objects, and structured data.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

## What the milestone delivers

- A catalog of 500 stable Oracle behavior contracts and 2,000 case specifications across ten exact domains.
- Bounded execution of 920 governed core SQL/type cases, 320 PL/SQL cases, 280 transaction and operations cases, and 480 schema/DML, schema-object, and structured-data cases.
- Schema and structured-data coverage spanning DML state changes, defaults, identity, constraints, indexes, schema evolution, views, sequences, synonyms, partitioning, materialized views, index-organized tables, editions, BLOB, CLOB, SecureFiles, JSON, XMLType, and object types.
- Separately declared expectations and independently computed observations across 100 topic families, five behavior focuses, and four case dimensions.
- A deduplicated evidence ladder reporting 500 catalogued, 500 catalog-executed, 500 unique bounded-model verified, 0 native-verified, and 0 target-equivalent behaviors, with 2,024 evidence records after retaining the 24 bootstrap runs separately.

- Native execution plans plus deterministic artifacts, ten JSON Schemas, cross-platform verification, and adversarial dependency-drift, mutation, overlap, and overclaim tests.

## How it differs from earlier work

MS #49 proved the mechanism with eight fixtures and 24 cases. MS #50.0 made them bootstrap evidence inside a governed broad-coverage program; MS #50.1 executes 920 core SQL/type cases, MS #50.2 executes 320 PL/SQL cases, MS #50.3 executes 280 transaction and operations cases, and MS #50.4 executes the final 480 schema/DML, schema-object, and structured-data cases without promoting bounded-model evidence into native Oracle proof.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

## What a customer can do with it

A customer can use oracle semantic coverage program as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

## Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

## Boundaries and claims not made

- The 2,000 passed catalog cases are deterministic bounded-model evidence; they do not constitute native Oracle execution or conformance.
- No governed catalog cases remain unexecuted, but native Oracle verified and target-equivalent behavior counts remain zero.
- The 24 MS #49 bootstrap runs remain separate evidence records and all eight behavior bindings overlap executed catalog tranches.
- Physical storage and index behavior, partitioning, materialized-view refresh, edition visibility, SecureFiles and LOB locator semantics, native JSON/XML/object behavior, and the 19c-to-26ai JSON datatype delta remain bounded until authorized native evidence exists.
- No native Oracle 19c or 26ai execution receipt exists.
- Catalogued is not supported; bounded-model verified is not native conformant; native verified is not target equivalent.
- iDempiere projection, CloudBank mapping, application equivalence, migration completion, and production readiness remain false.

## Source of record

- CHANGELOG.md release section(s): 0.50.0, 0.50.1, 0.50.2, 0.50.3, 0.50.4
- README.md product and operating guidance

- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity